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[23TCSE301]

THIRD SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

OPERATING SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5x2=10**		
1.	**a.**	Define operating system?	Remember	CO1	**02**
	b.	What is the purpose of the Banker's algorithm?	Apply	CO2	**02**
	c.	Give the merits of paging in memory management.	Remember	CO3	**02**
	d.	Outline the role of device drivers.	Understand	CO4	**02**
	e.	Distinguish between SSDs over HDDs.	Understand	CO4	**02**
		PART-B	**6X5=30**		
2.	**a.**	List and explain the different types of operating systems with example.	Understand	CO1	**05**
		OR			
	b.	What are the core functionalities of an operating system?	Understand	CO1	**05**
3.	**a.**	Define turnaround time and waiting time in the context of process scheduling.	Understand	CO2	**05**
		OR			
	b.	State the conditions for a deadlock to occur.	Understand	CO2	**05**
4.	**a.**	Differentiate between binary semaphore and counting semaphore.	Understand	CO2	**05**
		OR			
	b.	What is the role of a page table in paging?	Understand	CO3	**05**
5.	**a.**	Explain the concept of segmentation in memory management.	Understand	CO3	**05**
		OR			
	b.	Differentiate between internal and external fragmentation.	Understand	CO3	**05**
6.	**a.**	Compare single-level and multi-level directory structures in terms of efficiency.	Analyze	CO4	**05**
		OR			
	b.	Explain the purpose of file attributes in a file system.	Analyze	CO4	**05**

7.	**a.**	Enumerate the process of memory allocation in the Linux operating system, highlighting the role of the Buddy-Heap algorithm.	Understand	C05	**05**
		OR			
	b.	Describe how Linux handles process scheduling, particularly focusing on the Time-Sharing and Real-Time scheduling algorithms.	Apply	CO5	**05**
		PART-C	**6X10=60**		
8.	**a.**	Explain the purpose and functionalities of system calls and how they provide an interface between the user programs and the operating system.	Understand	CO1	**10**
		OR			
	b.	Distinguish between hard real time systems and soft real time systems.	Understand	CO1	**10**
9.	**a.**	Calculate the average turnaround time (TAT) and waiting time (WT) for the following set of processes using the FCFS scheduling algorithm. **Process** **Arrival Time** **Burst Time** P1 0 4 P2 1 3 P3 2 1	Apply	CO2	**10**
		OR			
	b.	Discuss the role of semaphores in solving the Producer-Consumer problem.	Apply	CO2	**10**
10.	**a.**	Explain the process of deadlock recovery and the various strategies used.	Understand	CO2	**10**
		OR			
	b.	Illustrate how the FIFO page replacement algorithm works with an example.	Analyze	CO3	**10**
11.	**a.**	Explain the concept of demand paging.	Analyze	CO3	**10**
		OR			
	b.	Compare and contrast paging and segmentation.	Analyze	CO3	**10**

12.	**a.**	Consider a disk which is having 16 platters and each platter is divided into two surfaces. Each surface is having 1k tracks. Each track is further divided into 512 sectors. Each sector can store 2 KB data. Calculate the capacity of the disk. How many bits are required to identify the specific sector of the hard disk?	Apply	CO4	**10**
		OR			
	b.	Explain the different disk scheduling algorithms.	Understand	CO4	**10**
13.	**a.**	Compare the key features and design principles of Linux, Android, and iOS operating systems.	Analyze	CO5	**10**
		OR			
	b.	Analyze the microkernel and monolithic kernel architectures of operating systems. Discuss their design principles, advantages, and disadvantages, and provide examples of operating systems that utilize each architecture.	Analyze	CO6	**10**